COURIER MANAGEMENT SYSTEM

Salesforce-Based Cloud Application

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1. INTRODUCTION AND PROJECT OVERVIEW

Courier and logistics companies can face operational difficulties when shipment tracking, delivery assignment, customer communication and billing are handled through manual or disconnected processes. The supplied project identifies delayed delivery updates, poor customer visibility, scheduling conflicts, missing automated notifications and limited performance analytics as key business problems.

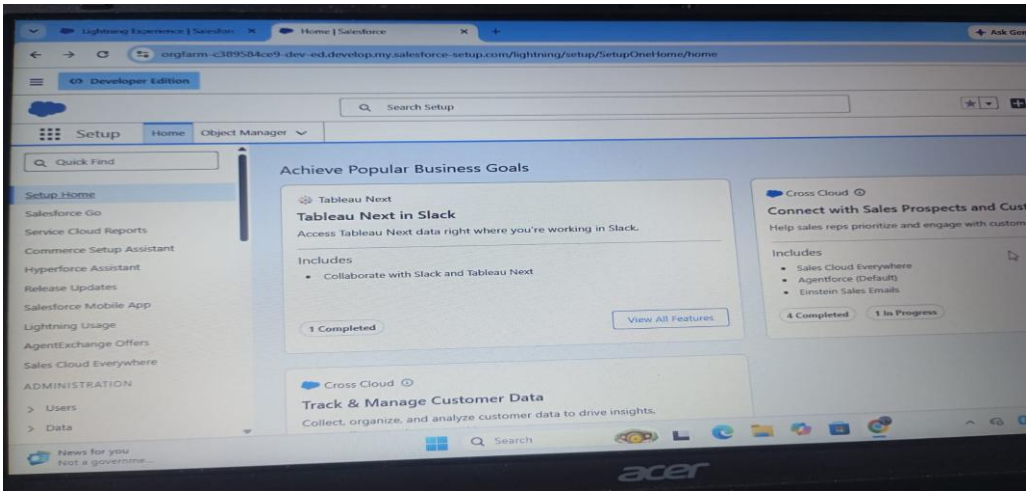
The Courier Management System (CMS) is proposed as a Salesforce-based cloud solution. It centralizes shipment booking, tracking, delivery-agent assignment, automated status updates, billing and analytics. The project therefore combines operational data and automation within one platform.

The application is intended to support multiple branches and provide a consistent way to manage courier records. The use of Salesforce Lightning gives users a common interface, while Salesforce Objects provide the underlying data structure.

The project is not limited to shipment tracking. It also covers customer management, branch management, deliveryagent information, delivery updates and invoices. Reports and dashboards provide a management view of operational performance.

Project goals

- Centralize courier information and reduce disconnected records.
- Improve visibility of shipment status and delivery progress.
- Automate repeated status-update operations.
- Protect data through profiles and role-based access.
- Provide reporting for shipment status, agent performance and revenue.



2. PURPOSE, PROBLEM STATEMENT AND IDEATION

2.1 Purpose

The purpose of CMS is to automate and centralize courier and logistics operations using Salesforce. The system is intended to replace manual and fragmented processes with integrated shipment booking, real-time tracking, delivery-agent assignment, automated notifications, billing and performance reporting.

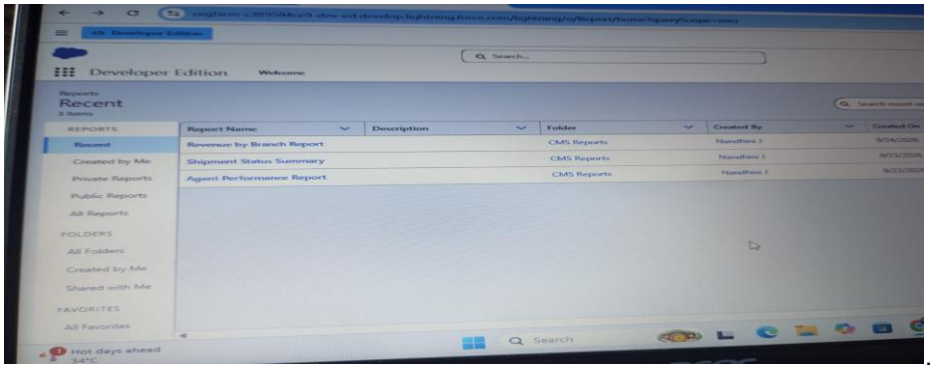
2.2 Problem Statement

User	Goal	Problem	Effect
Branch Manager	Manage shipments efficiently	Tracking is manual and systems are disconnected	Stress and operational difficulty
Customer	Track parcels	Updates are delayed and there is no centralized system	Uncertainty about delivery progress

2.3 Ideation

The project considered manual registers, spreadsheet tracking, a standalone portal and Salesforce CRM. Salesforce CRM was selected in the source document because of its automation capability, scalability and real-time visibility.

The ideation phase therefore moves from identifying operational pain points to selecting a platform capable of



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3. EMPATHY MAP AND CUSTOMER JOURNEY

3.1 Empathy Map – Branch Manager

Area	Observation
Think & Feel	Concerned about delays and errors
Hear	Customer complaints
See	Registers and spreadsheets
Say & Do	Manual follow-ups
Pain	Delivery conflicts and missed updates
Gain	Automated tracking and dashboards

3.2 Customer Journey Map

The supplied journey follows the courier process from booking to final reporting. Each stage represents a point where data is created or updated.

Stage	Main Activity
1	Booking
2	Shipment Creation
3	Agent Assignment
4	Transit
5	Delivery Update
6	Invoice
7	Reporting

This sequence provides the foundation for the later object relationships, automation flow and reports.

4. REQUIREMENTS ANALYSIS

4.1 Functional Requirements

ID	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates

FR-5	Invoice generation
FR-6	Reports and dashboards

4.2 Non-Functional Requirements

Requirement	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

4.3 Data Flow

> *Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports*

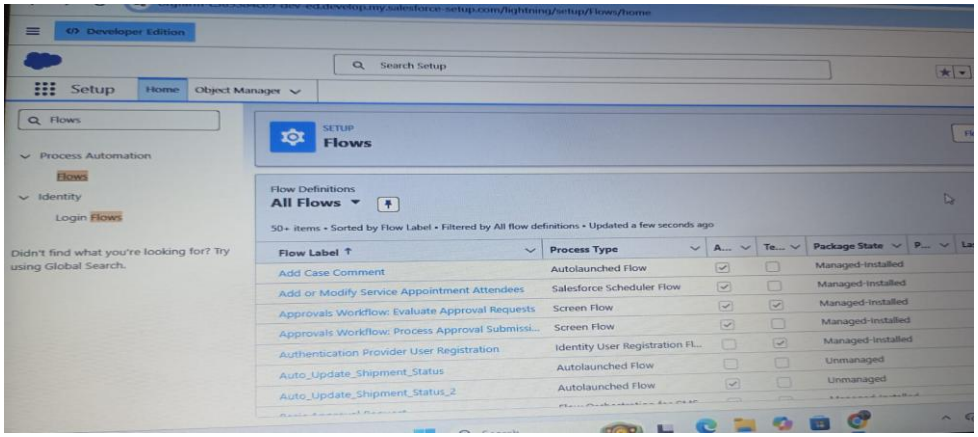
The data flow shows how operational records move through the system and eventually become management information.

5. TECHNOLOGY STACK AND DATA FLOW

The project uses Salesforce as the central application platform. Each technology layer corresponds to a specific responsibility in the CMS architecture.

Layer	Technology / Feature	Purpose
User Interface	Salesforce Lightning	User interaction and navigation
Business Logic	Salesforce Flows	Automation and record updates
Database	Salesforce Objects	Storage of courier data
Automation	Validation Rules and Flows	Data quality and process automation
Security	Profiles and Roles	Permissions and data visibility
Reporting	Reports and Dashboards	Operational analysis

The combination allows the application to be configured as an integrated cloud solution. The source document emphasizes that Salesforce Objects act as the database layer, while Lightning, Flow and reporting features provide the application and analytics layers.



Data moves through related records rather than isolated files. For example, a Shipment can be linked to a Customer and a Delivery Agent, while Delivery Updates are connected to the Shipment and invoices reference shipments for billing.

6. PROJECT DESIGN AND PROPOSED SOLUTION

6.1 Problem–Solution Fit

The project addresses fragmented courier operations by bringing shipment tracking, delivery assignment, billing and reporting into a centralized Salesforce platform.

6.2 Proposed Solution

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce-based Courier Management System
Innovation	Automated tracking and billing
Impact	Faster delivery operations
Business Model	Courier services
Scalability	Multi-branch ready

6.3 Design Principles

- Use related Salesforce objects to keep operational data connected.
- Use validation rules to prevent invalid or incomplete records.
- Use Flow to automate status synchronization.

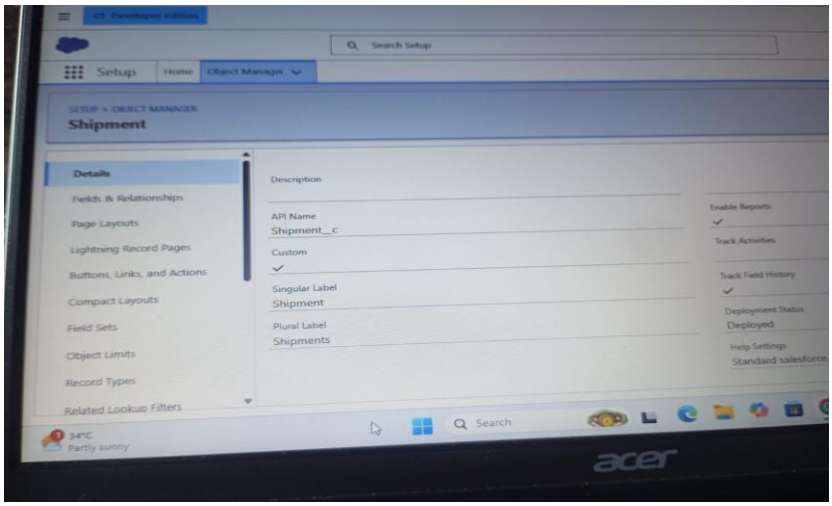
- Use profiles and roles to separate responsibilities.
- Use reports and dashboards to convert operational records into management information.

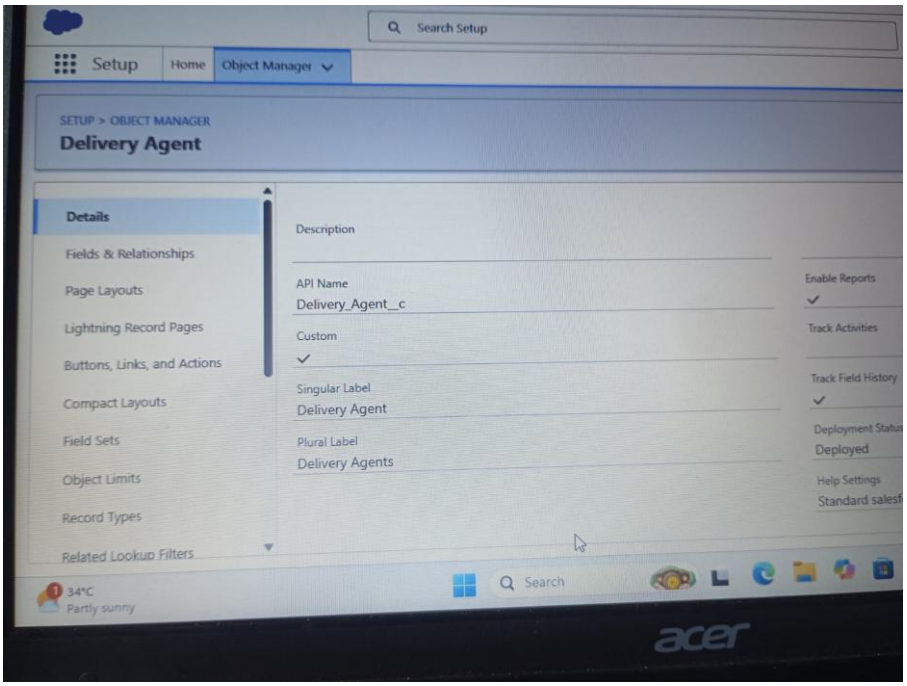
7. SOLUTION ARCHITECTURE

The solution architecture is organized around six major custom objects and the Salesforce services that operate on them.

Component	Role in the System
Customer__c	Stores customer personal and contact information
Shipment__c	Stores parcel, source, destination and tracking information
Delivery_Agent__c	Stores delivery-agent profiles and branch information
Branch__c	Represents courier branch offices
Delivery_Update__c	Stores status updates for shipments
Invoice__c	Manages billing details for delivered shipments
Flows	Automate record updates
Reports	Summarize operational information
Dashboards	Present visual management information

The architecture is intentionally modular. Each object has a focused responsibility, while relationships connect the records into an end-to-end courier process.




Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Delivery Agent
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Description
API Name
Delivery_Agent__c
Custom
Singular Label
Delivery Agent
Plural Label
Delivery Agents
Enable Reports
Track Activities
Track Field History
Deployment Status
Deployed
Help Settings
Standard salesf
34°C
Partly sunny
Search
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8. PROJECT PLANNING AND SCHEDULING

The supplied project follows an Agile methodology using epics, user stories, story points and sprint-based execution.

Velocity is mentioned as an estimation technique.

Sprint	Epic / User Story	Points	Priority
1	Developer setup – create and configure Salesforce account	3	High
2	Data modeling – create Customer, Shipment, Delivery Agent, Branch and Invoice objects	5	High
2	Create tabs and Lightning app for courier modules	5	High
3	Automation – validate shipment details	3	High
3	Automation – status updates and agent assignment	3	High
4	Security – role-based access control	5	High
5	Reports – delivery performance, shipment status and revenue	4	High
6	Dashboards – delivery performance, shipment status and revenue	4	Medium

The project tracker in the source also records Sprint-1 through Sprint-4 with six-day durations and planned release dates. The planning structure supports incremental implementation rather than building every feature at once.

9.SALESFORCE ACCOUNT AND OBJECT MODEL

9.1 Salesforce Account

The project begins with creation of a Salesforce Developer account. The supplied instructions specify the Salesforce Developer signup page and ask for basic registration information such as name, email, job title, company/college name and country. Account activation is completed through the verification email, followed by password and security-question setup.

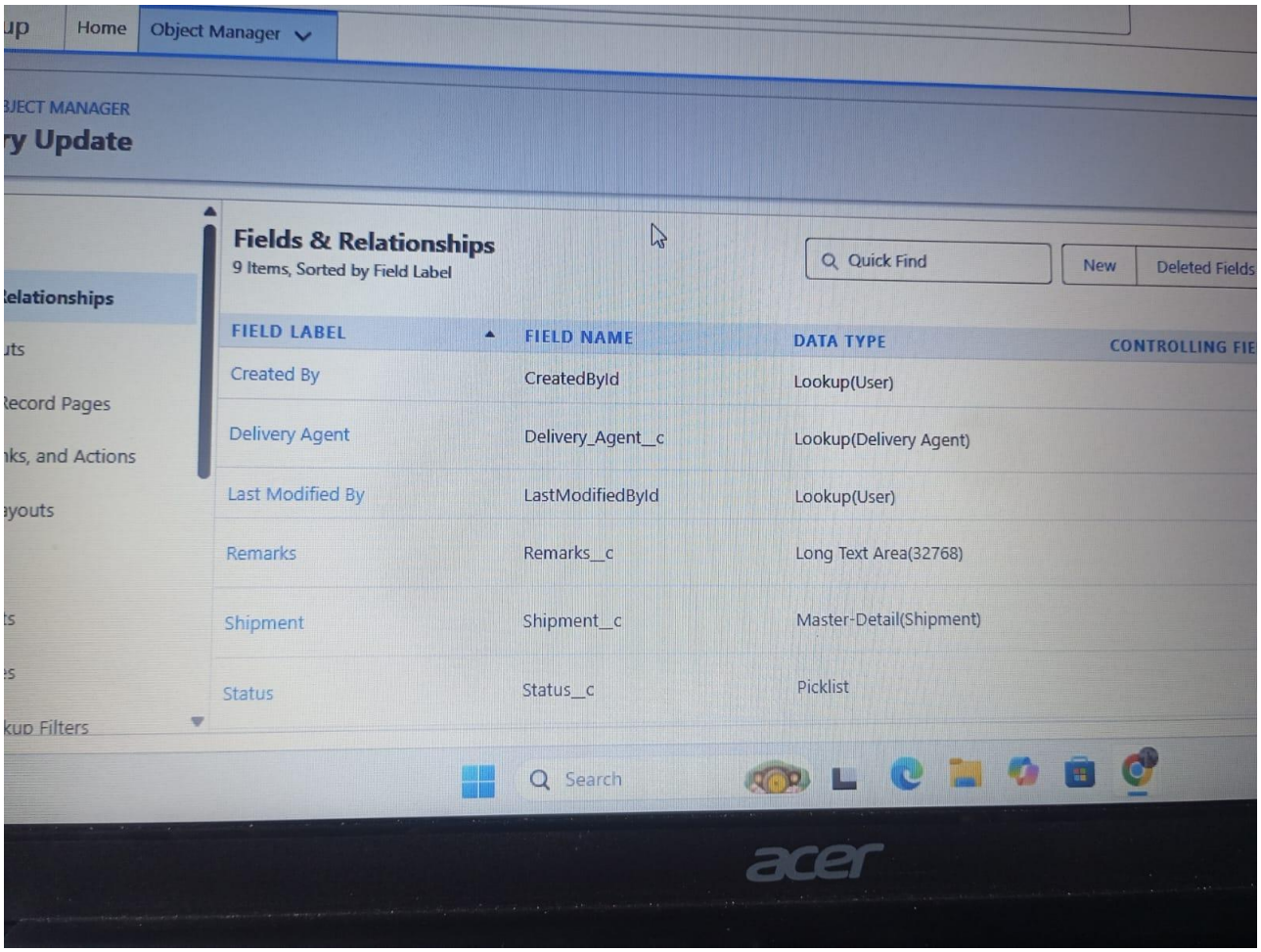
9.2 What is a Salesforce Object?

The source defines Salesforce objects as database tables that store organization-specific data. Fields represent columns and records represent rows.

9.3 Custom Objects

Object	Primary Purpose
Customer__c	Customer information
Shipment__c	Parcel and tracking information
Delivery_Agent__c	Delivery agent information
Branch__c	Branch office information
Delivery_Update__c	Shipment status history
Invoice__c	Billing information

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hese objects form the database foundation of the CMS.

10. CUSTOM OBJECTS AND TABS

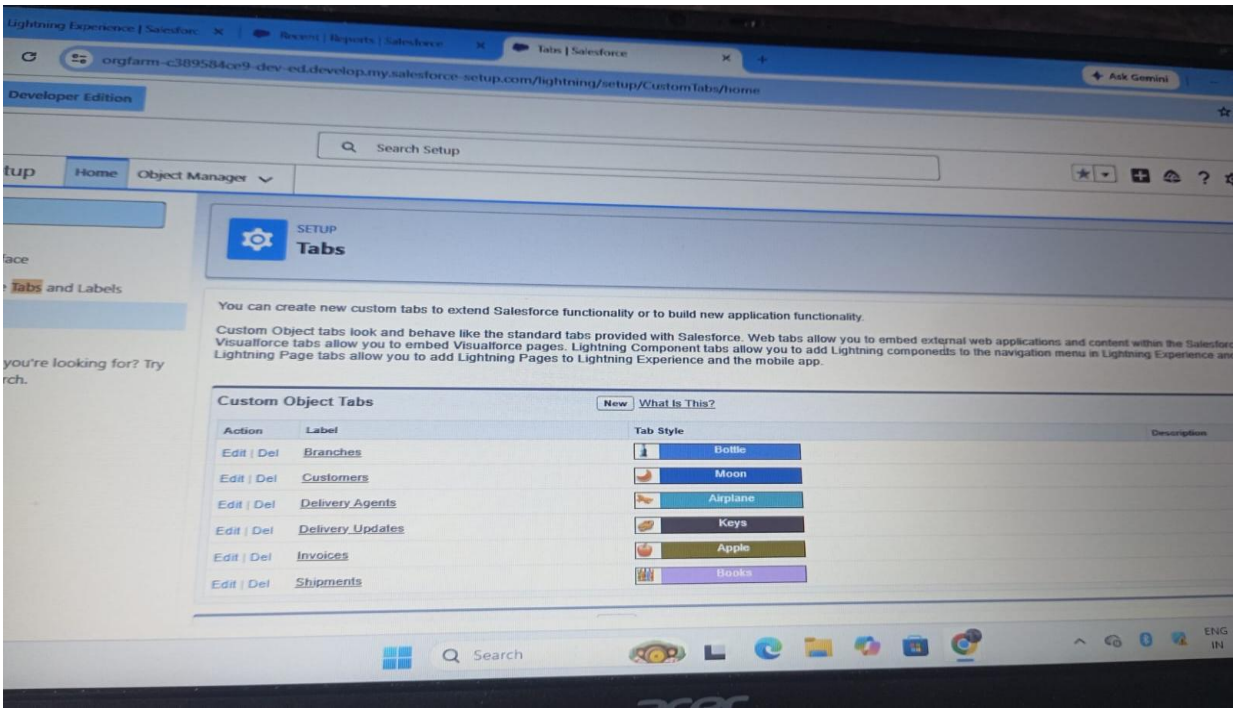
10.1 Object Configuration Summary

Object	Record Name	Key Configuration
Customer	Customer Name (Text)	Reports, field history, search, notes/attachments
Shipment	Tracking Number (Auto Number)	Format SHP-{00000}; reports and field history
Delivery Agent	Agent Name (Text)	Reports and field history
Branch	Branch Name (Text)	Reports
Delivery Update	Update Number (Auto Number)	Format UPD-{0000}; reports and field history
Invoice	Invoice Number (Auto Number)	Format INV-{0000}; reports and field history

10.2 Custom Tabs

The project creates custom tabs for Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice. The supplied procedure uses Setup → Tabs → Custom Object Tabs → New, followed by selecting the object and a tab style.

These tabs make the custom objects directly accessible from the Lightning application and support daily navigation.

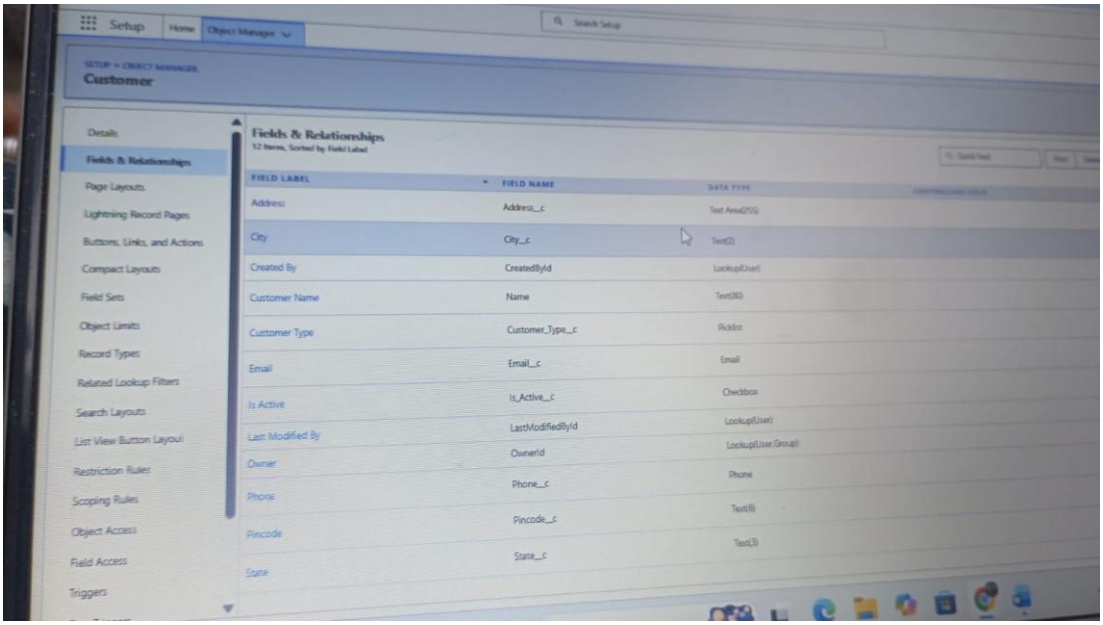


11. CUSTOMER AND SHIPMENT FIELDS

11.1 Customer Fields

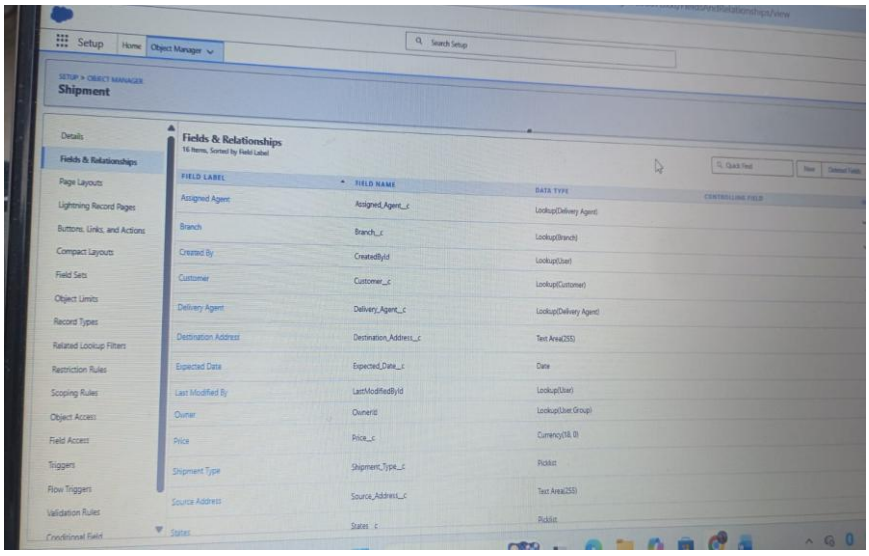
Field	API Name	Type	Required
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes
Customer Type	Customer_Type__c	Picklist	Yes
City	City__c	Text	No
State	State__c	Text	No
Pincode	Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

Customer Type values are Regular and Business. The source also specifies Email, Phone and Address as required fields.



11.2 Shipment Fields

Field	Type	Required
Tracking Number	Auto Number	Yes
Customer	Lookup(Customer)	Yes
Source Address	Text Area	Yes
Destination Address	Text Area	Yes
Weight	Number	Yes
Price	Currency	Yes
Status	Picklist	Yes
Expected Date	Date	Yes
Assigned Agent	Lookup(Delivery Agent)	No
Shipment Type	Picklist	Yes



12.1 Delivery Agent

Field	API Name	Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

Agent status values: Active, Inactive and On Leave.

12.2 Branch

Field	API Name	Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

12.3 Delivery Update and Invoice

Object	Important Fields
Delivery Update	Update Number, Shipment, Status, Update Date, Remarks
Invoice	Invoice Number, Shipment, Amount, Invoice Date, Payment Status

13. RELATIONSHIPS AND DATA MODEL

The relationships connect the six major objects into one operational model.

Child Object	Field	Parent Object	Relationship
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery Agent	Lookup
Delivery Agent	Branch__c	Branch	Lookup
Delivery Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Relationship Interpretation

- A customer can be associated with shipments through the Customer lookup.
- A shipment can have an assigned delivery agent.
- A delivery agent is associated with a branch.
- Delivery updates are master-detail children of a shipment, allowing status history to remain connected to theshipment.
- Invoices reference shipments so billing information can be related to the delivery record.

The model supports the customer journey from booking through delivery and billing while keeping the records connected for reporting.

14. VALIDATION RULES

Validation rules check data when a record is saved. If the condition evaluates to true, Salesforce prevents the record from being saved and displays an error message.

Object	Rule	Formula / Condition	Error
Customer	Email Required	ISBLANK(Email__c)	Email is required.
Customer	Phone Length	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Customer	Pincode Length	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Shipment	Weight	Weight__c <= 0	Weight must be greater than zero.
Shipment	Price	Price__c <= 0	Price must be greater than zero.
Shipment	Expected Date	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Agent	Email	NOT(CONTAINS(Email__c,"@"))	Enter a valid email address.

Branch	Branch Code	ISBLANK(Branch_Code__c)	Branch Code is mandatory.
Delivery Update	Delayed Remarks	Delayed + blank Remarks	Remarks are required when status is Delayed.

These rules improve data quality before records enter the reporting and automation processes.

15. PAGE LAYOUTS

Page layouts control which fields, sections, buttons and related lists users see on record pages. The source organizes each object into meaningful sections.

Object	Sections / Main Fields
Customer	Customer Information; Address Details
Shipment	Shipment Details; Address Information; Delivery Information
Delivery Agent	Agent Details; Additional Information
Branch	Branch Information
Delivery Update	Update Information
Invoice	Invoice Details

Customer layout example

The Customer Information section contains Customer Name, Customer Type, Email, Phone and Is Active. Address Details contains Address, City, State and Pincode.

Shipment layout example

Shipment Details contains Tracking Number, Customer, Shipment Type, Weight, Price and Status. Address Information contains Source Address and Destination Address. Delivery Information contains Expected Date and Assigned Agent.

Consistent layouts reduce user confusion and make data entry easier.

16. SALESFORCE FLOW AUTOMATION

16.1 Scenario

Whenever a Delivery_Update__c record is created, the related Shipment__c Status should automatically be updated with the latest status from the Delivery Update.

16.2 Flow Configuration

Step	Configuration
Flow Type	Record-Triggered Flow
Object	Delivery_Update__c
Trigger	A record is created
Conditions	None – always run
Optimization	Actions and Related Records
Update Element	Update_Shipment_Status
Target Object	Shipment__c
Filter	Shipment__c Id equals {!$Record.Shipment__c}
Field Updated	Shipment Status
Value Source	{!$Record.Status__c}

16.3 Result

The flow copies the status from the newly created Delivery Update to the linked Shipment. This removes the need for a separate manual update and keeps shipment status synchronized with delivery activity.

The final implementation step is to save the flow with the label Auto_Update_Shipment_Status and activate it.

17. LIGHTNING APPLICATION

17.1 Purpose of the App

A Salesforce app is a group of tabs that work together as a unit. The project creates a dedicated Courier Management System Lightning App so that users can access the courier modules from one place.

17.2 App Configuration

Configuration	Value
App Type	New Lightning App
App Name	Courier Management System
Main Tabs	Customer, Shipment, Delivery Agent, Branch
Operational Tabs	Delivery Update, Invoice
Analytics Tabs	Reports, Dashboards
Profile	System Administrator

17.3 Navigation

The application brings operational records and analytics together. A user can move from customer and shipment information to delivery updates and invoices, then access reports and dashboards for monitoring.

The source implementation creates the app through Setup → App Manager → New Lightning App, adds the required items and assigns the selected profile.

18. PROFILES AND SECURITY

Profiles define what users can do in Salesforce, including object permissions, field permissions, user permissions, tab settings, app settings and related access controls.

Profile	Purpose	Main Access
Courier Admin	Full system control	Full access inherited from System Administrator base
Branch Manager	Manage branch operations	Customer, Shipment, Agent, Branch, Update and Invoice access as specified
Delivery Agent	Update delivery status	Shipment and Delivery Update access; limited access to other areas
Customer Support	Handle customers and tracking	Customer, Shipment, Delivery Update and Invoice read access as specified

The Branch Manager profile is based on Standard Platform User and receives access to the Courier Management System App and the specified tabs. Delivery Agent and Customer Support profiles are also based on standard profiles and are configured with more limited access.

The source also specifies session and password settings for selected profiles, including a two-hour inactivity timeout and an eight-character minimum password length.

19. ROLES AND USERS

19.1 Role Hierarchy

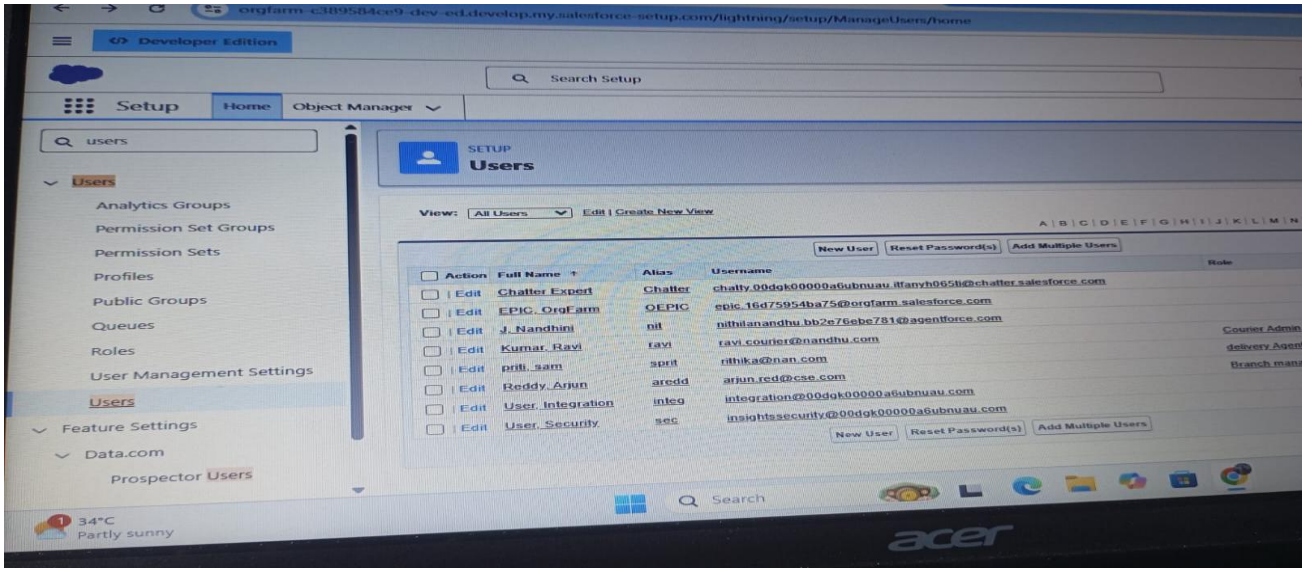
Level	Role	Purpose
1	Courier Admin	Highest level access to all data
2	Branch Manager	Manages branch-level operations
3	Delivery Agent	Works with assigned shipment and delivery information
3	Customer Support	Handles customer and shipment tracking

The hierarchy places Branch Manager below Courier Admin. Delivery Agent and Customer Support are placed below Branch Manager. This structure is used to support reporting hierarchy and data visibility.

19.2 Users

Salesforce users are the people who log in to the system. The source specifies that each user should have a unique username, valid email, assigned role and assigned profile.

User Type	Role	License	Profile
Courier Admin	Courier Admin	Salesforce	Courier Admin
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support



20. REPORTS AND DASHBOARDS

Reports and dashboards provide the analytics layer of the Courier Management System. The source defines custom report types and three main reports.

20.1 Report Types

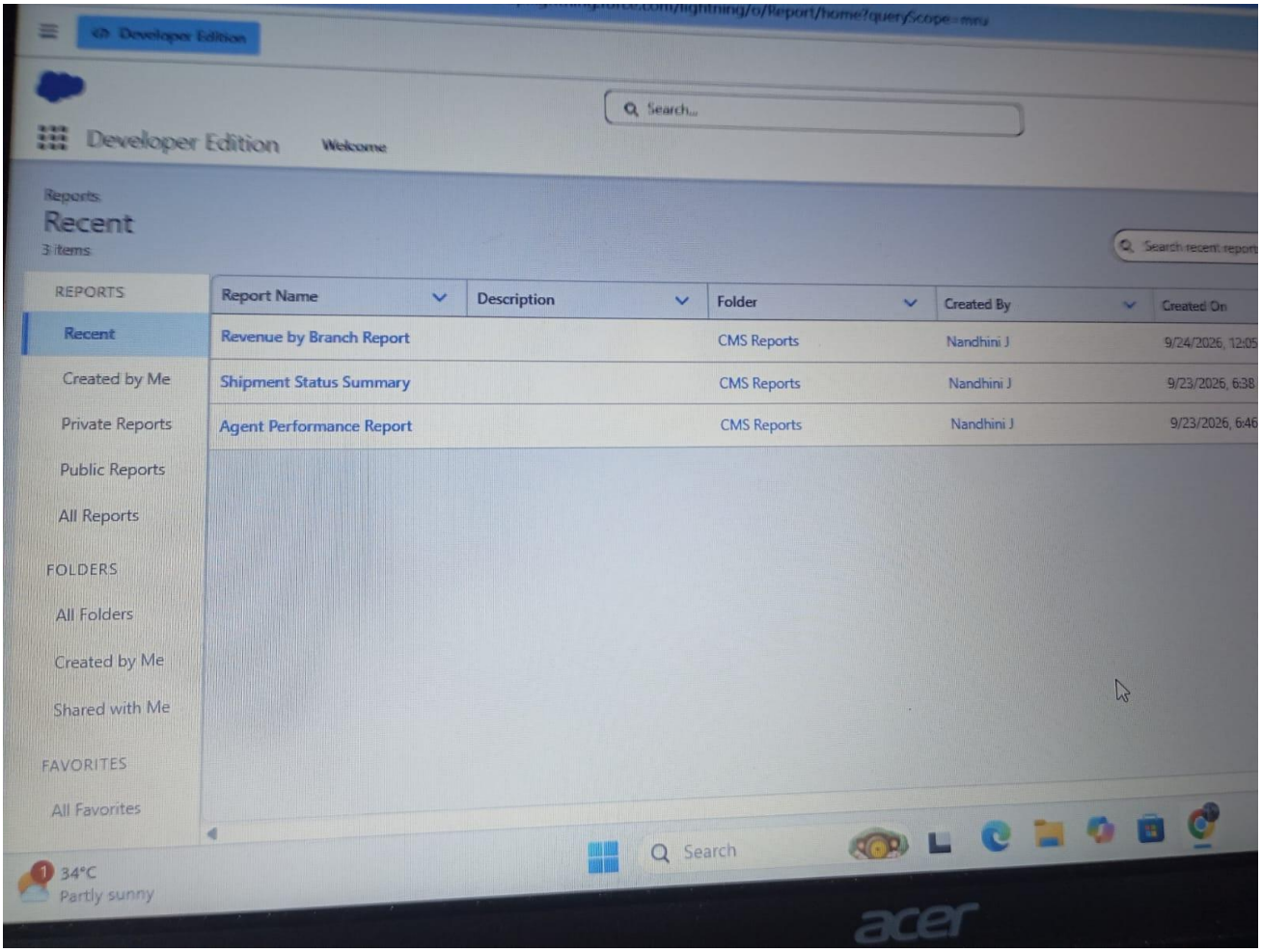
Report Type	Primary Object	Label
Shipment	Shipment__c	Shipment Tracking Report
Customer	Customer__c	Customer Summary Report
Agent	Delivery_Agent__c	Agent Performance Report

20.2 Reports

Report	Purpose / Main Fields
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent; grouped by Status with Price summary
Agent Performance Report	Agent Name, Branch, Number of Shipments, Status; grouped by Delivery Agent

Revenue by Branch Report	Tracking Number, Customer, Price, Invoice Status; grouped by Branch and Month

The reports are stored in the CMS Reports folder in the supplied configuration.



21. DASHBOARD COMPONENTS

The Courier Management Dashboard provides a visual summary of shipment status, agent performance and revenue.

Component	Source Report	Chart Type
Shipment Status Overview	Shipment Status Summary	Donut Chart
Agent Performance	Agent Performance Report	Bar Chart
Revenue by Branch	Revenue by Branch Report	Column Chart

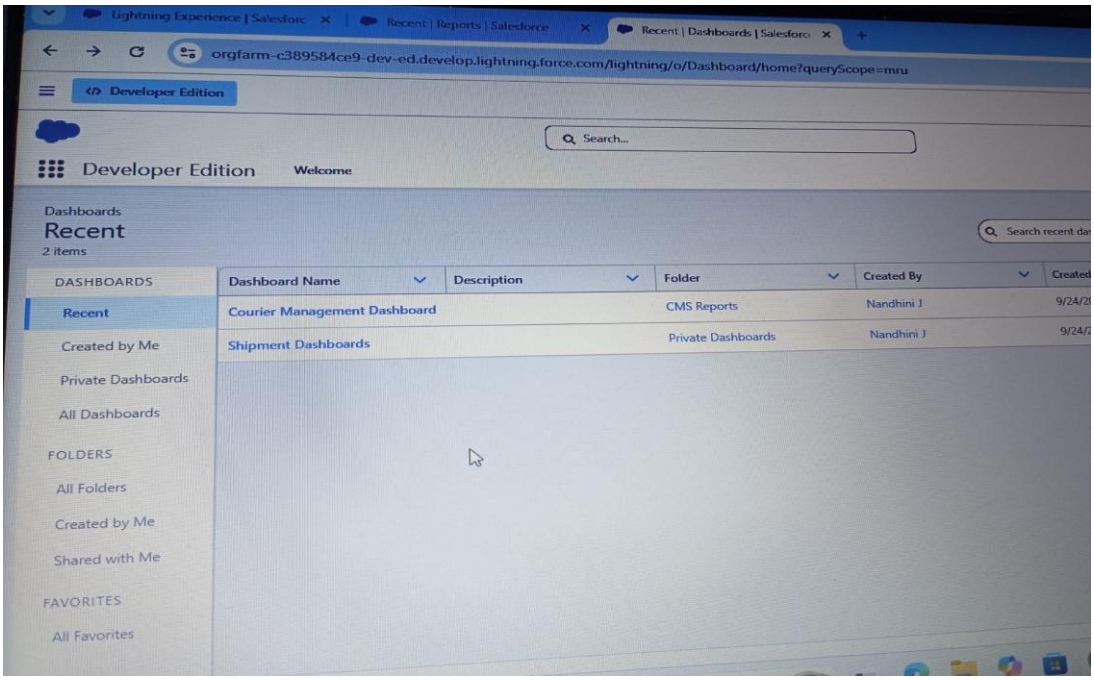
Dashboard Creation

The supplied procedure creates a dashboard named Courier Management Dashboard in the CMS Reports folder. Components are added using the dashboard component option and are connected to the previously created reports.

Dashboard Sharing

The source specifies sharing the dashboard with Courier Admin, Branch Managers and the Customer Support Team.

The dashboard therefore acts as the management-facing view of the system, while the underlying reports provide the detailed data.



22. FUNCTIONAL AND PERFORMANCE TESTING

The project includes testing to verify that the configured Salesforce application behaves as intended. The source states that screenshots were captured from Salesforce to verify record creation, validation-rule enforcement, delivery status updates, flow automation, invoice generation and dashboard visualization.

22.1 Test Areas

Test Area	Expected Verification
Record Creation	Customer, shipment, agent, branch, update and invoice records can be created
Validation	Invalid or incomplete data is rejected with configured error messages
Relationships	Related records point to the correct parent objects
Flow Automation	Delivery Update creation changes related Shipment status
Invoice	Billing information is stored against a shipment
Reports	Required operational data appears in report outputs
Dashboard	Charts display shipment, agent and revenue information
Security	Profiles and roles provide the configured access boundaries

23. ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

23.1 Advantages

- Centralized data: courier information is stored within a connected Salesforce environment.
- Automated workflows: Salesforce Flow reduces repetitive manual updates.
- Real-time reporting: reports and dashboards provide operational visibility.
- Role-based access: profiles and roles separate responsibilities.
- Multi-branch readiness: the Branch object and hierarchy support branch-oriented operations.

23.2 Limitations

- Salesforce dependency: the solution is built on the Salesforce platform.
- Initial configuration effort: objects, fields, relationships, validation, security and reporting require setup.

23.3 Future Scope

- Mobile tracking application.
- GPS integration.

24. CONCLUSION AND LEARNING OUTCOMES

24.1 Conclusion

The Courier Management System digitizes courier operations using Salesforce automation. The project connects customers, shipments, delivery agents, branches, delivery updates and invoices through a structured object model. Validation rules improve data quality, while Salesforce Flow automates shipment-status synchronization.

The Lightning application provides a common interface for daily operations, and profiles and roles establish a structured access model. Reports and dashboards convert operational data into management information for shipment status, agent performance and revenue.

24.2 Learning Outcomes

Learning Area	Outcome
Salesforce Setup	Create and configure a Developer org
Data Modeling	Design custom objects and relationships
UI Configuration	Create tabs, page layouts and a Lightning app
Validation	Implement rules for data quality
Automation	Build record-triggered flows
Security	Configure profiles, roles and users
Analytics	Create reports and dashboards
Testing	Validate end-to-end application behavior

25. IMPLEMENTATION CHECKLIST AND APPENDIX

25.1 Implementation Checklist

- Salesforce Developer account created and activated.
- Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice objects created.
- Custom tabs created for all major objects.
- Required fields and picklists configured.
- Lookup and master-detail relationships configured.
- Validation rules created and activated.
- Page layouts organized into logical sections.
- Auto_Update_Shipment_Status flow created and activated.
- Courier Management System Lightning App created.

- Courier Admin, Branch Manager, Delivery Agent and Customer Support profiles configured.
- Role hierarchy created.
- Required users created with roles and profiles.
- Custom report types and operational reports created.
- Courier Management Dashboard created with three components.
- Functional and performance testing completed.

25.2 Appendix

Source Code / Configuration: Salesforce Configuration

Dataset: Salesforce Objects

Demo / Development Environment: Salesforce Developer Org

The supplied source identifies the project implementation primarily through Salesforce configuration rather than a separate traditional source-code repository.

26. FINAL PROJECT SUMMARY

Courier Management System is a Salesforce-based cloud application designed to bring courier operations into a centralized platform. The project starts with business problems such as manual tracking, delayed updates, disconnected systems and limited operational analytics.

The solution addresses these issues through a connected data model consisting of Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice objects. Lookup and master-detail relationships connect the records, while validation rules protect data quality.

Automation is provided through a record-triggered Salesforce Flow that copies a Delivery Update status to its related Shipment. A dedicated Lightning application brings the modules together for users. Profiles, roles and users establish structured access and reporting hierarchy.

Finally, custom report types, operational reports and a Courier Management Dashboard provide management visibility into shipment status, agent performance and revenue by branch.

Area	Final Deliverable
Business Analysis	Problem statement, empathy map, customer journey and requirements
Design	Architecture, data model and project plan
Development	Custom objects, fields, relationships and layouts
Automation	Validation rules and shipment-status Flow

Security	Profiles, roles and users
Analytics	Reports and Courier Management Dashboard
Quality	Functional and performance testing
Future	Mobile, GPS, SMS and AI-based extensions

